Day 4 Mastery Quiz: Comparisons and Boundaries — Instructor Key

Converted value → comparison operator → exact boundary → type-safe threshold note

Instructor key. Same layout as student copy; solutions filled in response boxes. Print duplex on one sheet.

Name		**Section**		**Date**	
Score	______/10		Mastery check	secure / developing / needs support	

The pump station has converted its intake fields into numeric flow values. A reading passes the single-value boundary check only when the operator and exact boundary are correct. Do not combine conditions yet; show the comparison evidence one line at a time.

1 Reference code for Items 1–10

predict trace

```
# converted values from the Day 3 intake process
target_lpm = 21.0
tolerance_lpm = 0.6
low_limit_lpm = target_lpm - tolerance_lpm
high_limit_lpm = target_lpm + tolerance_lpm
reading_low_edge = 20.4
reading_mid = 21.0
reading_high_edge = 21.6
reading_over = 21.61
reading_text = "21.6"

# individual comparison checks
reading_low_edge >= low_limit_lpm
reading_low_edge > low_limit_lpm
reading_high_edge <= high_limit_lpm
reading_high_edge < high_limit_lpm
reading_over <= high_limit_lpm
reading_mid == target_lpm
```

2 Items 1–5: exact-boundary predictions

predict explain

Pt.	Prompt	My response
1	What values are stored in `low_limit_lpm` and `high_limit_lpm`?	**Key:** `low_limit_lpm = 20.4` and `high_limit_lpm = 21.6`.
1	What is the result of `reading_low_edge >= low_limit_lpm`, and why?	**Key:** `True`; exact lower boundary is included by `>=`.
1	What is the result of `reading_low_edge > low_limit_lpm`, and why?	**Key:** `False`; equal is not greater than.
1	What is the result of `reading_high_edge <= high_limit_lpm`, and why?	**Key:** `True`; exact upper boundary is included by `<=`.
1	What is the result of `reading_high_edge < high_limit_lpm`, and why?	**Key:** `False`; equal is not less than.

3 Items 6–8: mismatch and type-transfer checks

debug test

Pt.	Prompt	My response
1	What is the result of `reading_over <= high_limit_lpm`, and which side of the boundary controls the result?	**Key:** `False`; `21.61` is above the upper boundary `21.6`.
1	What is the result of `reading_mid == target_lpm`? Explain why `==` is not assignment.	**Key:** `True`; `==` compares equality and does not store a value.
1	Why is `reading_text <= high_limit_lpm` not a safe boundary check in Python?	**Key:** `reading_text` is a string and `high_limit_lpm` is a float; cast before comparing.

4 Item 9: debug a boundary note

debug explain

Boundary note to check

The `20.4` reading fails because it is not greater than the lower limit, and the `21.6` reading fails because it is not less than the upper limit.

Pt.	Prompt	My response
1	Rewrite the note using inclusive boundary language for both exact-limit readings.	**Key:** The `20.4` lower-edge reading passes with `>=` and the `21.6` upper-edge reading passes with `<=`; exact limits are included.

5 Item 10: prepare, but do not combine, the Day 5 evidence

implement transfer

Pt.	Prompt	My response
1	Write the two separate comparison expressions that should be checked for `reading_high_edge`. Do not connect them with `and`.	**Key:** `reading_high_edge >= low_limit_lpm` and `reading_high_edge <= high_limit_lpm`.

Support route, not scored

My issue is mostly: setup / Python syntax / type conversion / equality vs assignment / inclusive boundary / exclusive boundary / confidence / time.
My next small action is: test just-below, exact, just-above / rewrite the operator / check type before comparison / ask a partner / ask instructor.